

Transform Techniques - Tutorial 1

1 Fourier series and the Fourier transform.

1. Find the period of the function $f(x) = \sin(\frac{2\pi nx}{k})$
2. Does a constant function have the fundamental period¹?
3. Let $f(x)$ be a function of period 2π given by

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ x, & 0 < x < \pi. \end{cases}$$

- a) Sketch a graph of $f(x)$ in the interval $-3\pi < x < 3\pi$.
- b) Find the Fourier series for $f(x)$ in the interval $-\pi < x < \pi$.
- c) By giving appropriate values to x , show that

$$(i) \quad \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots,$$

$$(ii) \quad \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots.$$

4. Assume f is a function on the interval $[0, \pi]$ whose graph is symmetric with respect to the line $x = \pi/2$; in other words,

$$f(\pi - x) = f(x) \quad (0 \leq x \leq \pi).$$

Suppose the Fourier sine series of f on $[0, \pi]$ is

$$f(x) \sim \sum_{k=1}^{\infty} b_k \sin(kx).$$

Prove that $b_k = 0$ for every even k .

5. Find the Fourier transforms of the following functions :

$$(a) \quad f(t) = \begin{cases} 0 & \text{if } t < 0 \\ e^{-\omega t} & \text{if } t > 0 \end{cases} \text{ where } \omega > 0.$$

$$(b) \quad f(t) = \begin{cases} 1 - t/2 & \text{if } 0 \leq t \leq 2 \\ 1 + t/2 & \text{if } -2 \leq t \leq 0 \\ 0 & \text{if } |t| > 2 \end{cases}$$

6. Suppose $\hat{f}$ denotes the Fourier transform of the function f .

(a) Is it true that if f is odd (even), then $\hat{f}$ is odd (even)?

(b) Let $g(t) = f(t - t_0)$. Then prove that $\hat{g}(\alpha) = e^{-i\alpha t_0} \hat{f}(\alpha)$

(c) Let $g(t) = e^{i\alpha t_0} f(t - t_0)$, then prove that $\hat{g}(\alpha) = \hat{f}(\alpha - \alpha_0)$

¹fundamental period is the smallest positive period.

7. Given that the Fourier transform of the function $f(t) = \begin{cases} 1/2 & \text{if } |t| \leq 1 \\ 0 & \text{if } |t| > 1 \end{cases}$ is $\frac{\sin(\alpha)}{\alpha}$. Find the inverse fourier transform of the function $L(\alpha) = \frac{\sin(6\alpha)}{3\alpha - \pi}$.